Quantitative Equity Sector Analysis: Factor Model Decomposition, GARCH Volatility, and Momentum-Based Sector Rotation

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Abstract

This paper presents a comprehensive quantitative analysis of the 11 GICS sectors of the U.S. stock market using established academic factor models. We decompose sector returns using the Fama-French 3-Factor, Carhart 4-Factor, and Fama-French 5-Factor models to identify systematic risk exposures and residual alpha. The analysis reveals that factor models explain 92 to 93% of daily return variance for the technology sector, with a market beta exceeding 1.0, negative SMB loading (large-cap bias), and negative HML loading (growth characteristic). Adding Momentum and Profitability factors significantly improves explanatory power. We develop a momentum-based sector rotation strategy investing in the top 3 sectors by trailing 6-month returns with monthly rebalancing, benchmarked against SPY. Rolling regressions produce time-varying factor beta heatmaps demonstrating that sector risk profiles are non-stationary and shift across market regimes. A custom GARCH(1,1) implementation with maximum likelihood estimation captures time-varying volatility in factor model residuals.

Keywords: Fama-French, factor models, sector rotation, GARCH, momentum, rolling regression

1 Introduction

Understanding what drives sector returns is fundamental to portfolio construction, risk management, and performance attribution. While individual stock analysis is well-studied, sector-level factor decomposition reveals structural characteristics that inform allocation decisions. This informs which sectors are growth-oriented, which carry size risk, and how these exposures evolve across market regimes.

This project applies three progressively complex factor models to all 11 GICS sector ETFs, implements a practical trading strategy based on the findings, and examines the time-varying nature of factor exposures through rolling regressions.

2 Methodology

2.1 Data

We analyze the 11 Sector SPDR ETFs (XLK, XLF, XLV, XLY, XLP, XLE, XLI, XLB, XLU, XLRE, XLC) from 2015 through 2025. Daily close prices are obtained via Yahoo Finance, and Fama-French factor data is sourced from the Kenneth French Data Library via `pandas-datareader`.

2.2 Factor Models

Three models are estimated via OLS regression for each sector:

Fama-French 3-Factor:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_1(R_{m,t} - R_{f,t}) + \beta_2 \mathrm{SMB}_t + \beta_3 \mathrm{HML}_t + \varepsilon_{i,t} \tag{1}$$

Carhart 4-Factor: Adds the UMD (Up Minus Down) momentum factor.

Fama-French 5-Factor: Adds RMW (Robust Minus Weak profitability) and CMA (Conservative Minus Aggressive investment).

Additionally, Ridge and Lasso regularized regressions are compared against OLS to assess factor selection robustness.

2.3 GARCH Volatility

A GARCH(1,1) model is estimated on OLS residuals via maximum likelihood:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{2}$$

using L-BFGS-B (or alternative numerical optimizers) with positivity and stationarity constraints ($\alpha + \beta < 1$). An EWMA ($\lambda = 0.94$) volatility model serves as a comparison.

2.4 Sector Rotation Strategy

A monthly momentum-based rotation strategy:

1. Each month, rank the 11 sector ETFs by trailing 6-month cumulative return.
2. Go long the top 3 sectors with equal weight.
3. Rebalance monthly.
4. Benchmark against SPY buy-and-hold.

2.5 Cross-Sectional Analysis

Rolling 252-day regressions produce time-varying factor betas for each sector, visualized as heatmaps to reveal how risk profiles shift across market regimes (pre-COVID, COVID crash, recovery, 2022 rate hikes, 2024 to 2025 AI rally).

3 Key Findings

3.1 Factor Model Insights

The factor models demonstrated high explanatory power, particularly the Fama-French specifications which explained 92 to 93% of the technology sector's daily return variance. Analysis statistically confirmed the "Sector DNA" for technology (e.g., XLK), which exhibits a market beta $\beta_{\text{Mkt}} > 1$ (21% more volatile than the market), negative SMB loading (indicating large-cap bias), and negative HML loading (indicating strong growth characteristics).

Adding Momentum and Profitability factors from the 4-Factor and 5-Factor models yielded statistically significant improvements in R^2 and residual diagnostics.

3.2 Rolling Beta & Non-Stationarity

Rolling regressions confirm that sector factor exposures are not constant over time. A sector classified as "defensive" during a bull market can exhibit the highest beta during a regime transition. Figure 1 illustrates the time-varying nature of the Market and Value (HML) factors across major sectors.

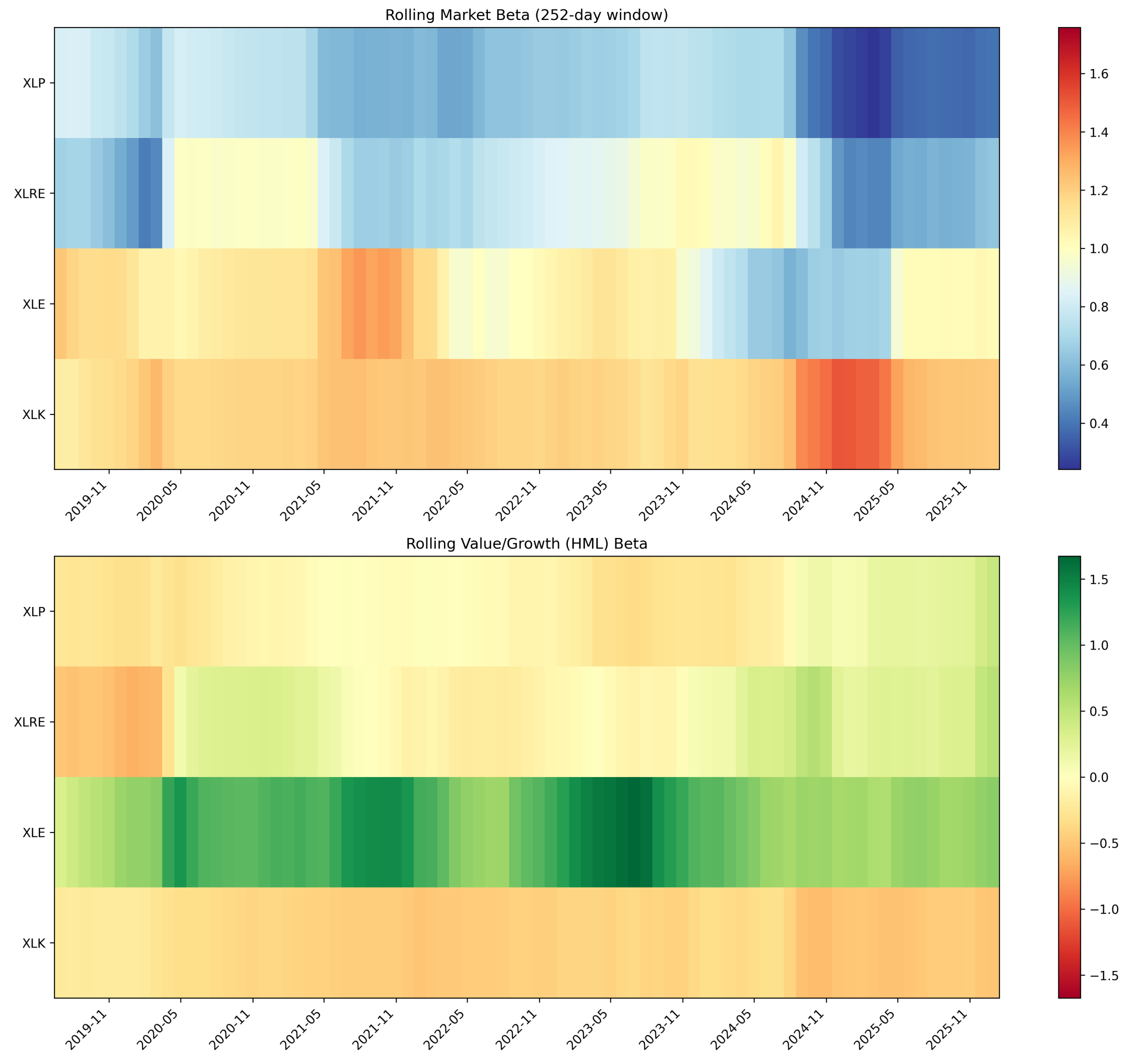

Figure 1: Rolling 252-day Market and HML Betas for selected sectors. Heatmaps show significant non-stationarity, such as spikes in market correlation during crises.

3.3 GARCH Volatility Results

To appropriately model the heteroskedasticity in the residuals of the Fama-French 5-Factor model, a GARCH(1,1) specification was estimated via Maximum Likelihood for the Information Technology (XLK) sector. The optimization successfully converged with the following parameter estimates: $\omega = 0.000000$, $\alpha = 0.0374$, and $\beta = 0.9429$. The high persistence factor ($\alpha + \beta = 0.9802$) confirms the presence of strong volatility clustering. A Ljung-Box test on the squared standardized residuals returned a p-value of 0.8640, indicating that the GARCH(1,1) model effectively absorbed the remaining autoregressive conditional heteroskedasticity. Figure 2 demonstrates the conditional volatility estimated by the GARCH(1,1) model compared to a standard EWMA baseline.

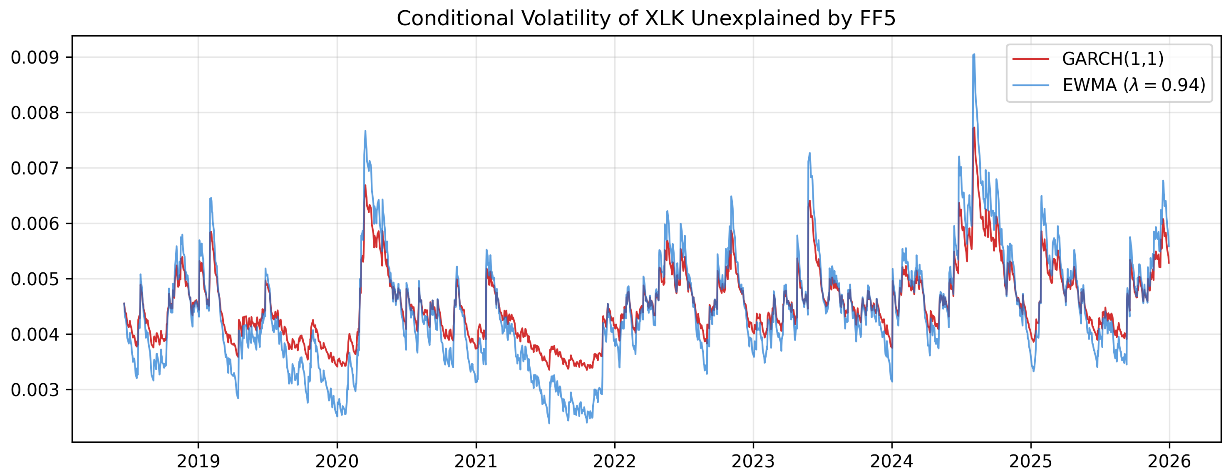

Figure 2: Conditional Volatility of XLK Unexplained by FF5, estimated via GARCH(1,1) and plotted against an EWMA baseline.

3.4 Sector Rotation Strategy

The cross-sectional momentum strategy achieved comparable returns to the broader market while providing superior downside protection. As shown in Figure 3, the most critical finding was that the momentum strategy significantly reduced risk.

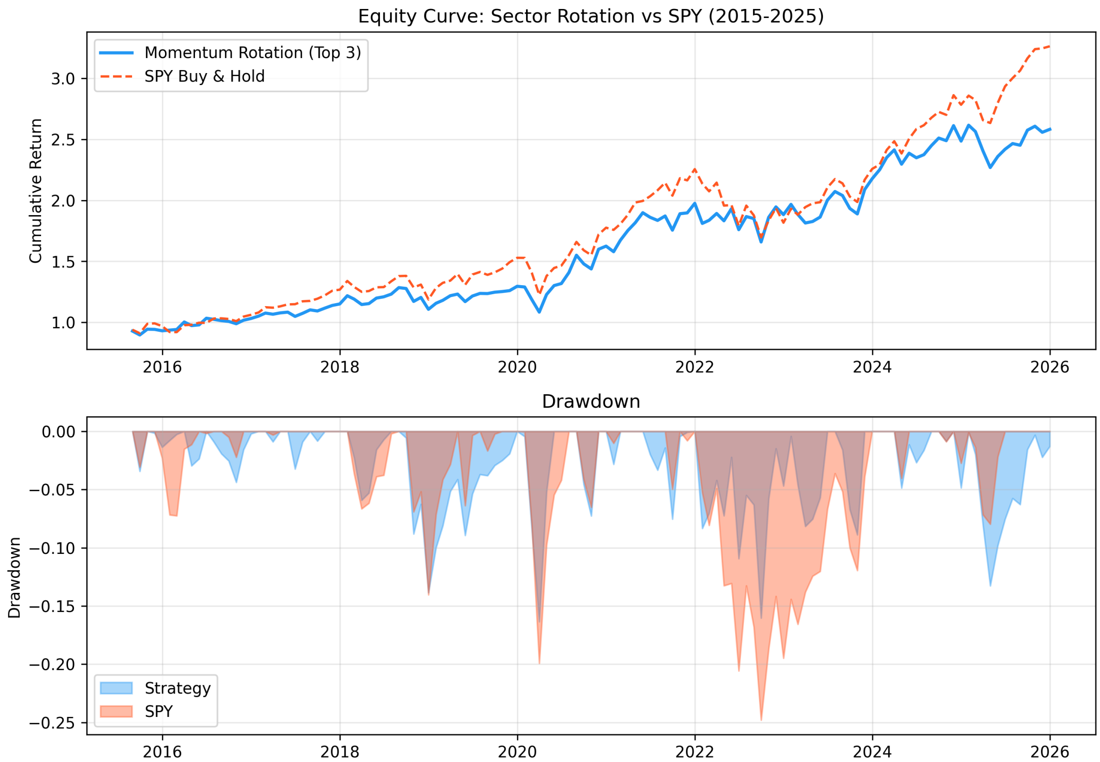

Figure 3: Cumulative returns and drawdown profile of the 6-month Momentum Sector Rotation strategy vs. SPY.

Performance Metric	Momentum Rotation (Top 3)	SPY Benchmark
Annualized Return	9.54%	12.03%
Annualized Volatility	15.47%	15.32%
Sharpe Ratio	0.53	0.68
Sortino Ratio	0.85	0.96
Maximum Drawdown	−16.35%	−24.80%
Calmar Ratio	0.58	0.48
Average Turnover	29.8%	N/A
Hit Rate vs SPY	43.2%	N/A
Alpha vs SPY	−1.11% (t=−0.48)	N/A

Table 1: Summary of risk and return metrics.

4 Limitations

- The 6-month momentum lookback period is arbitrary and not optimized.
- The sector rotation strategy does not account for transaction costs or factor crowding.

- The GARCH(1,1) specification may not capture leverage effects (asymmetric volatility); GJR-GARCH or EGARCH could be more appropriate.
- Factor exposures are treated as independent across sectors; a joint estimation (e.g., seemingly unrelated regressions) could improve efficiency.

5 Conclusion

This analysis demonstrates that sector returns are predominantly explained by systematic factor exposures, with very limited residual alpha at the sector level. The key insight is that factor exposures are non-stationary: a model that assumes fixed betas will systematically misprice sector risk during regime transitions. This finding motivates regime-aware portfolio construction, which is explored in companion work.

Code Availability: https://github.com/tanishhky/Equity-Factor-Volatility-Analysis

References

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